

FPT = (fixed-parameter tractable) algorithms

n = input size k = some relevant parameter of the input

cost: $O(\underbrace{f(k)} n^{O(1)})$ time where $f(k)$ can grow very fast
but only in k (not in n ...)

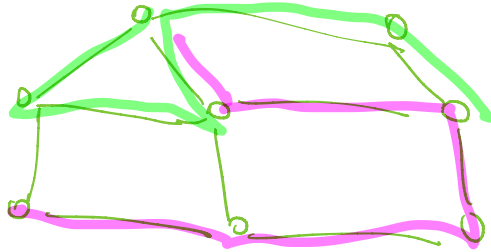
Randomization in FPT algorithms (Monte Carlo)

↳ color coding ✓

↳ randomized separation (related to 2-coloring) ✓

COLOR CODING

Running example: k -path = path of k nodes in a graph G ,
all nodes traversed once ($\neq$ walk)



$k=6$

— walk

— k -path

walk is easy

k -path is NP-hard ($k = \# \text{ nodes in } G \Rightarrow \text{HAM}(G)?$)

IDEA Random k -coloring: for each node u in G , assign color $X(u) \in [k]$
to u in a random and uniform fashion.

Scenario: $G=(V,E)$, set $[k]$ of color, random assignment $X:V \rightarrow [k]$

$$\Pr(X(u)=c) = \frac{1}{k} \quad \text{for any } X, u, c$$

Subset $V' \subseteq V$ is **COLORFUL** iff $\forall u \neq u' \text{ in } V': X(u) \neq X(u')$

(V' = set of nodes in a given k -path)

prop. Take a colorful walk of k nodes is a k -path

The vice versa is not true: in a k -path, there could be two nodes with the same color. However...

 obs. a walk is a k -path iff. $\exists$ coloring X s.t. the walk is colorful

Prop: $V' \subseteq V$ such that $|V'| = k$.

$$\Pr(V' \text{ is COLORFUL}) = \frac{\# \text{ colorings } \chi \text{ making } V' \text{ colorful}}{\# \text{ colorings } \chi}$$

each node in $V \setminus V'$
can take one out of k
colors, $|V \setminus V'| = n - k$

$$= \frac{k^{n-k} \cdot k!}{k^n}$$

each coloring of V'
should assign different colors
to its nodes

each of the $n - k$ nodes
can take one out of k colors

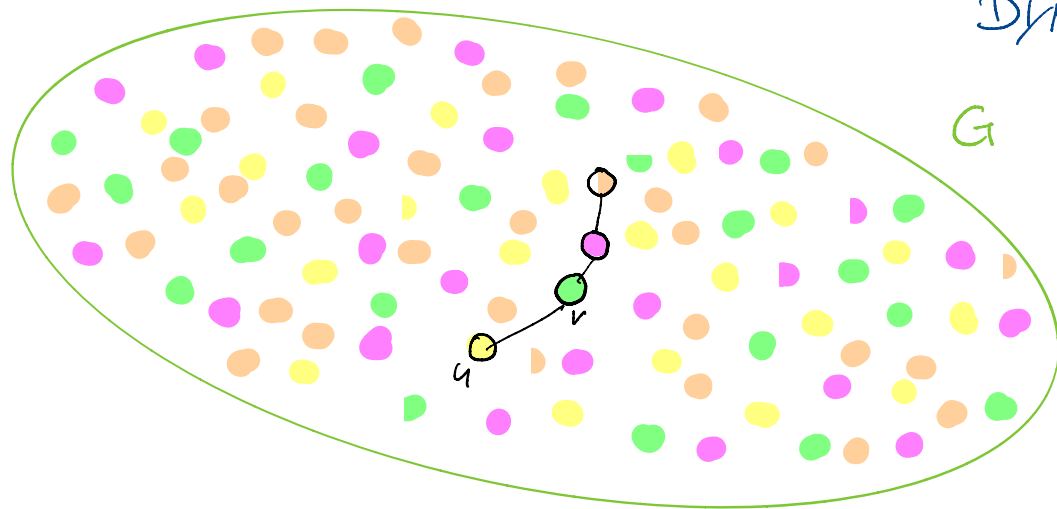
$$= \frac{k!}{k^k} > \frac{1}{e^k}$$

$$k! > \left(\frac{k}{e}\right)^k$$

Hence, V' is colorful with $\Pr > e^{-k}$
 $k = |V'|$

Given G, k, X , we want to discover all colourful paths (# nodes is $\leq k$)

DYNAMIC PROGRAMMING



$k=4$

obs 1 any subpath of a colourful path is clearly colourful

obs 2. A colourful path of l nodes ends in u iff

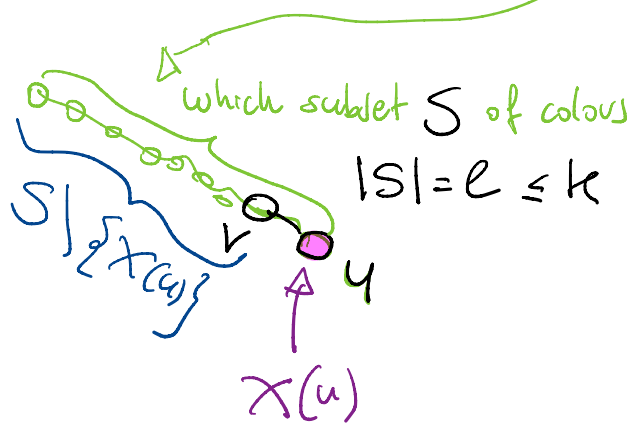
$\exists v \in N(u)$ s.t. a colourful path of $l-1$ nodes ends in v and
none of the latter nodes has colour $X(u)$

obs 3 The set of $l-1$ colours matters, not their order

obs 1-3 are crucial to get a FPT-alg. as we need to store, for each node, just its $\binom{k}{\ell}$ possible subsets of ℓ colors out of k

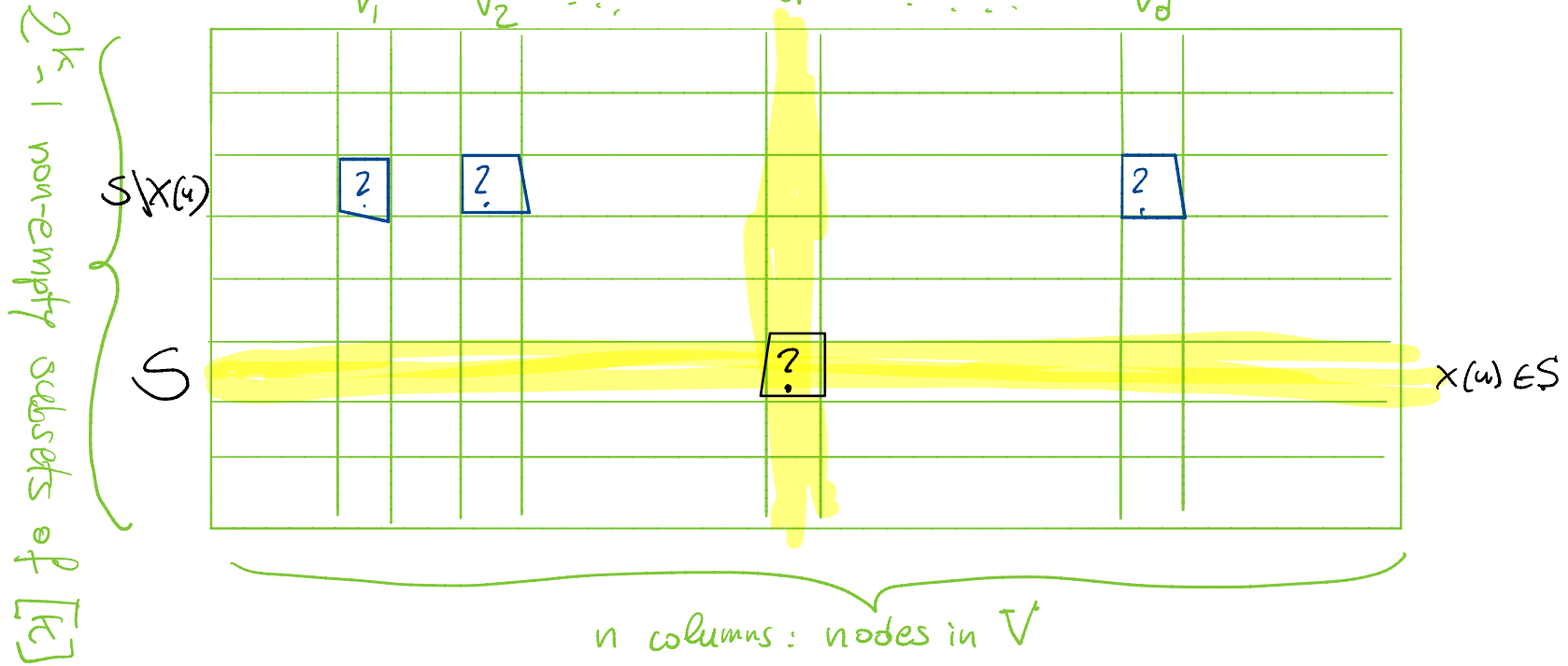
$$\hookrightarrow \sum_{\ell} \binom{k}{\ell} = 2^k$$

instead of storing the $O(n^k)$ paths ending in that node



$$\text{PATH}[S][u] = \begin{cases} T & \text{if } S = \{X(u)\} \\ \text{OR}_{v \in N(u)} \text{PATH}[S \setminus X(u)][v] & \text{if } X(u) \notin S, |S| \geq 2 \\ F & \text{otherwise} \end{cases}$$

$$N(u) = \{v_1, v_2, \dots, v_d\}$$



As $\text{PATH}[i]$ has $2^k - 1$ rows and $n = |V|$ columns, and each entry can be filled in $O(\underbrace{N(u)}_{\in n})$ time (for node u), total time is FPT, that is,
 $O(2^k n^2)$ time (actually, $O(2^k \cdot m)$)

(*) ob) There is a colorful k -path iff the last row in $\text{PATH}[i]$ contains an entry $T \Rightarrow pr > e^{-k}$ as seen before

(*) + (*) $\Rightarrow$ perform e^k random colorings (instead of k^n) and we find a k -path (if it exists) with constant probability

Total time = $O(e^k 2^k n^2) = O((2e)^k \cdot n^2)$ FPT

We can boost the probability using Chernoff bounds
 (It can be derandomized using hashing)

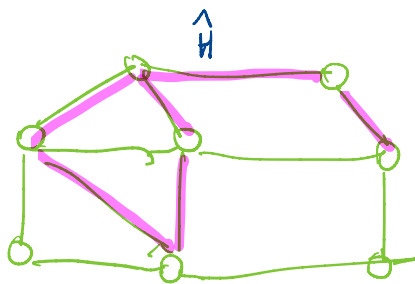
RANDOMIZED SEPARATION

Running example: SUBGRAPH ISOMORPHISM (includes k -path)

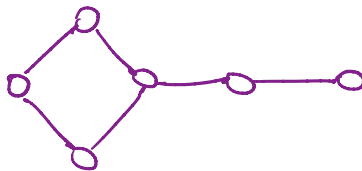
IN: $G = (V, E)$ $H = (V_H, E_H)$

OUT: $\hat{H} = (\hat{V}, \hat{E})$ subgraph of G : $\hat{V} \subseteq V, \hat{E} \subseteq E$ isomorphic to H

$\hat{H}$ isomorphic to H iff $\exists$ 1-1 mapping $\varphi: V_H \rightarrow \hat{V}$
s.t. $xy \in E_H \Rightarrow \varphi(x)\varphi(y) \in \hat{E}$



G



H

Let's parameterize the problem...

- $k = |V_H| \Rightarrow$ no FPT alg. for this choice of k unless $P = NP$
- $k = |V_H|$, $\Delta = \max \text{ degree in } G \Rightarrow$ FPT alg. $O(\underbrace{2^{\Delta \cdot k} \cdot k!}_{f(\Delta, k)} n^{O(1)})$ time
 $\nwarrow$ bounded degree

IDEA Suppose $\hat{H}$ subgraph of G and isomorphic to H exists

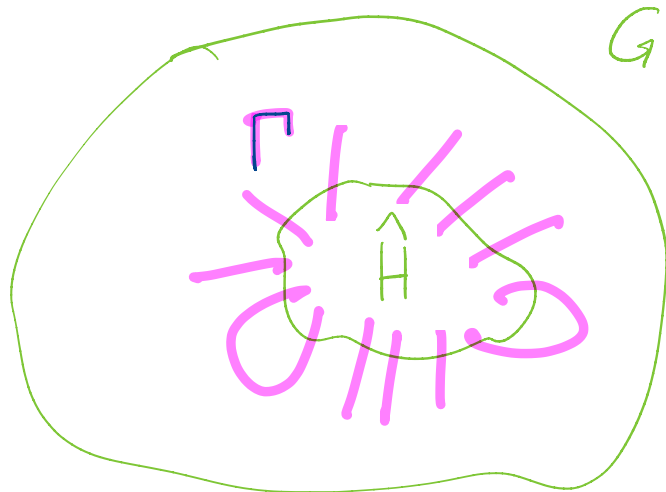
$$\Gamma = \{ xy \in E \setminus \hat{E} : x \in \hat{V} \text{ or } y \in \hat{V} \}$$

$\nwarrow$ rationale: Γ disconnects $\hat{H}$ from the rest of G

⊛ prop

$$\boxed{|\Gamma| + |\hat{E}| \leq |\hat{V}| \cdot \overset{\text{max degree}}{\Delta} = |V_H| \cdot \Delta = k\Delta}$$

$\swarrow$ disjoint by def.



Use random 2-coloring of edges to detach $\hat{H}$ from the rest of G

$X: E \rightarrow \{R, B\}$ is **successful** if:

(a) all the edges in $\hat{E}$ are the same color $c \in \{R, B\}$, and

(b) all the edges in Γ are the opposite color $\bar{c} \in \{R, B\} \setminus \{c\}$

We implemented our rationale for Γ : looking at the same-color edges in G , we observe that $\hat{H}$ is an isolated connected component

$$\Pr(X \text{ is successful}) \geq \frac{1}{2^{|\Gamma| + |\hat{E}|}} \geq \frac{1}{2^{\Delta k}} \quad (\textcircled{x})$$

FPT algorithm:

1] Preliminary check if G has max degree Δ

2] Repeat $2^{\Delta k}$ times:

2.a random 2-coloring of the edges in E

2.b G_R = part of G induced by edges of color R

2.c G_B = " " " " " " " B

2.d for each connected component CC of G_R and G_B :

if $|CC| = k$ and CC is isomorphic to H

then $\hat{H} = CC$ and stop

Cost of performing this check: go through all $k!$ 1-1 mappings and, for each, check if their $k \times k$ adjacency matrix is equal: $O(k! k^2)$ time

Total time: $O(2^{\Delta k} \cdot (k! k^2) \cdot n) = O(2^{\Delta k} \cdot k! \cdot n^{O(1)})$ FPT? with constant prob.

